Review Test Submission: Section 9: MLR qualitative predictors - indicator variables part 1

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Course	2021-ST311 Applied Statistics I
Test	Section 9: MLR qualitative predictors - indicator variables part 1
Started	15/11/20 11:08
Submitted	15/11/20 11:16
Due Date	18/11/20 12:00
Status	Completed
Attempt Score	23 out of 23 points
Time Elapsed	7 minutes
Instructions	

Deadline to earn full score :

Please ensure all open attempts are submitted at that time.

Instructions :

Take care when typing answers to "fill in blank" questions:
If numeric answers require rounding, input answers rounding to decimal places as instructed after answer box.
Input a text answer using one of the options provided after answer space. Ensure your spelling is correct and leave only one space between words.

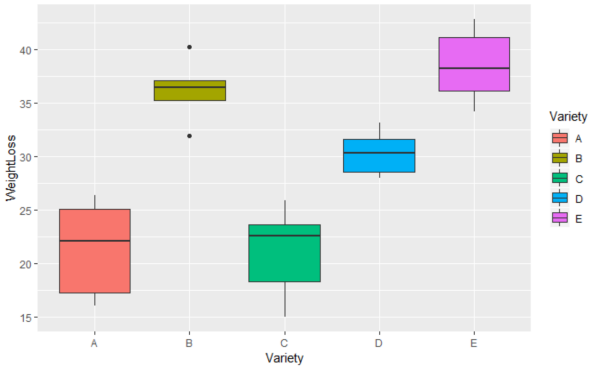
Results Displayed	Submitted Answers, Feedback, Incorrectly Answered Questions

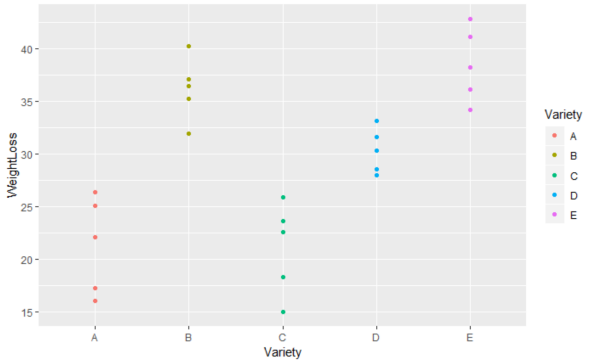
Question 1

7 out of 7 points

The owner of an orchard stores groups of apples in a number of different ways in his storage room. He wishes to assess the impact of these storage solutions on the percentage weight loss of the apples during storage. The percentage weight loss of 25 apples after storage with two columns of data as follows

- percentage weight loss of apple during storage
- group/storage method of apple A,B,C,D,E





Investigate if there is a significant difference in the average percentage weight loss of the different groups of apples during storage.

a) An indicator/dummy variable for group/storage method A is defined as:

IndA ¡=1 if weight loss observed for apple which is stored under method A

IndA ¡=0 otherwise.

Similar indicator/dummy variables can be defined for groups B to E, IndA, IndB, IndC, IndD, IndE. How many dummy variables do you need to fit a qualitative variable of 5 categories, if fitting using a regression model with a constant coefficient?

[a]

b) A model containing four such indicator variables IndA, IndB, IndC, IndD :

WeightLoss ¡=β_0 +β_1IndA ¡+β_2IndB ¡+β_3IndC ¡+β_4IndD ¡ + ε ¡

is fitted to the observed data giving the following output:

```
Call:  lm(formula = WeightLoss ~ IndA + IndB + IndC + IndD, data = apples)

Residuals:     Min      1Q Median      3Q     Max
            -6.056 -2.374  0.300  2.616  5.016

Coefficients:
            Estimate Std. Error   t value   Pr(>|t|)
(Intercept)   38.534      1.630     23.65      0.000
IndA         -17.130      2.305     -7.43      0.000
IndB          -2.314      2.305     -1.00      0.327
IndC         -17.428      2.305     -7.56      0.000
IndD          -8.174      2.305     -3.55      0.002
---
Residual standard error: 3.64401 on 20 degrees of freedom
Multiple R-squared:  0.832, Adjusted R-squared:  0.799
F-statistic: 24.81 on 4 and 20 DF,  p-value: 0.0000001645
```

Using the regression fit find:

An estimate of the average WeightLoss for apples stored under method A = **[avA]**. *(3 dec places)*.
An estimate of the average WeightLoss for apples stored under method B = **[avB]**. *(3 dec places)*.
An estimate of the average WeightLoss for apples stored under method C = **[avC]**. *(3 dec places)*.
An estimate of the average WeightLoss for apples stored under method D = **[avD]**. *(3 dec places)*.
An estimate of the average WeightLoss for apples stored under method E = **[avE]**. *(3 dec places)*.

c) Write down the meaning of the fitted constant term i.e. fitted value of β0

The value 38.534 is the estimated average WeightLoss for apples stored under method **[beta0]** *(Insert one of A/B/C/D/E)*.

Specified Answer for: a 4

Correct! A factor with 5 levels A,B,C,D,E needs only 4 dummy/indicator variables.

Specified Answer for: avA 21.404

Correct!

Specified Answer for: avB 36.22

Correct!

Specified Answer for: avC 21.106

Correct!

Specified Answer for: avD 30.36

Correct!

Specified Answer for: avE 38.534

Correct!

Specified Answer for: beta0 E

Correct!

Question 2

4 out of 4 points

Write down the interpretation of the fitted coefficient of IndA, i.e. the fitted value of β_1 and test the significance of this term for the population at significance level α=0.05.
Check your answer by filling in the blanks in the following:
Interpretation: The average WeightLoss for apples stored under condition A is **[intpt1]** *(greater than/less than)* the average WeightLoss for apples stored under condition **[intpt2]** by an estimated difference of **[intpt3]**. *(3 dec places)*
Test of significance: Testing $H_0: \beta_1 = 0$ versus $H_a: \beta_1 \neq 0$ the p-value is **[pval]** *(as printed)*.
There **[dec]** *(is evidence/is no evidence)* to suggest IndA is a good predictor of WeightLoss,
i.e. there **[dec2]** *(is evidence/is no evidence)* to suggest a significant difference in average WeightLoss for the population of apples stored under condition **[gp1]**

compared to the population of apples stored under condition **[gp2]**.

Specified Answer for: intpt1	less than
Specified Answer for: intpt2	E
Specified Answer for: intpt3	17.130
Specified Answer for: pval	0.00
Specified Answer for: dec	is evidence
Specified Answer for: dec2	is evidence
Specified Answer for: gp1	A
Specified Answer for: gp2	E

Question 3

4 out of 4 points

Write down the interpretation of the fitted coefficient of IndB, i.e. the fitted value of β_2 and test the significance of this term for the population at significance level α=0.05.
Check your answer by filling in the blanks in the following:
Interpretation: The average WeightLoss for apples stored under condition B is **[intpt1]** *(greater than/less than)* the average WeightLoss for apples stored under condition **[intpt2]** by an estimated difference of **[intpt3]**. *(3 dec places)*
Test of significance: Testing $H_0: \beta_2 = 0$ versus $H_a: \beta_2 \neq 0$ the p-value is **[pval]** *(as printed)*.
There **[dec]** *(is evidence/is no evidence)* to suggest IndB is a good predictor of WeightLoss,
i.e. there **[dec2]** *(is evidence/is no evidence)* to suggest a significant difference in average WeightLoss for the population of apples stored under condition **[gp1]** compared to the population of apples stored under condition **[gp2]**.

Specified Answer for: intpt1	less than
Specified Answer for: intpt2	E
Specified Answer for: intpt3	2.314
Specified Answer for: pval	0.327
Specified Answer for: dec	is no evidence
Specified Answer for: dec2	is no evidence
Specified Answer for: gp1	B
Specified Answer for: gp2	E

Question 4

4 out of 4 points

Write down the interpretation of the fitted coefficient of IndC, i.e. the fitted value of β_3 and test the significance of this term for the population at significance level α=0.05.
Check your answer by filling in the blanks in the following:
Interpretation: The average WeightLoss for apples stored under condition C is

[intpt1] *(greater than/less than)* the average WeightLoss for apples stored under condition **[intpt2]** by an estimated difference of **[intpt3]**. *(3 dec places)*
Test of significance: Testing $H_0: \beta_3 = 0$ versus $H_a: \beta_3 \neq 0$ the p-value is **[pval]** *(as printed)*.
There **[dec]** *(is evidence/is no evidence)* to suggest IndC is a good predictor of WeightLoss,
i.e. there **[dec2]** *(is evidence/is no evidence)* to suggest a significant difference in average WeightLoss for the population of apples stored under condition **[gp1]** compared to the population of apples stored under condition **[gp2]**.

Specified Answer for: intpt1	less than
Specified Answer for: intpt2	E
Specified Answer for: intpt3	17.428
Specified Answer for: pval	0.000
Specified Answer for: dec	is evidence
Specified Answer for: dec2	is evidence
Specified Answer for: gp1	C
Specified Answer for: gp2	E

Question 5

4 out of 4 points

Write down the interpretation of the fitted coefficient of IndD, i.e. the fitted value of β_3 and test the significance of this term for the population at significance level α=0.05.
Check your answer by filling in the blanks in the following:
Interpretation: The average WeightLoss for apples stored under condition D is **[intpt1]** *(greater than/less than)* the average WeightLoss for apples stored under condition **[intpt2]** by an estimated difference of **[intpt3]**. *(3 dec places)*
Test of significance: Testing $H_0: \beta_4 = 0$ versus $H_a: \beta_4 \neq 0$ the p-value is **[pval]** *(as printed)*.
There **[dec]** *(is evidence/is no evidence)* to suggest IndD is a good predictor of WeightLoss,
i.e. there **[dec2]** *(is evidence/is no evidence)* to suggest a significant difference in average WeightLoss for the population of apples stored under condition **[gp1]** compared to the population of apples stored under condition **[gp2]**.

Specified Answer for: intpt1	less than
Specified Answer for: intpt2	E
Specified Answer for: intpt3	8.174
Specified Answer for: pval	0.002
Specified Answer for: dec	is evidence
Specified Answer for: dec2	is evidence
Specified Answer for: gp1	D
Specified Answer for: gp2	E

There is no content to display.

Monday, 4 January 2021 14:14:20 o'clock GMT

← OK